

T.Y.B.Sc.(C.S.) NEP – 2020 Sem V Practical Examination

Lab Course CS-308-MJ-P

Data Science and Analytics

Instructions

1. Download the required datasets by using following reference dataset links and provide to the students at the time of the practical examination.
2. A reference dataset link is provided below the respective question. If the given link does not work, use an appropriate dataset relevant to the question.
3. Give the required input values wherever necessary, as specified in the question (e.g., $k = 3$ for K-Means Clustering or $k = 5$ for KNN).
4. Install the required Python libraries which ever necessary, such as pandas, numpy, scikit-learn, matplotlib, squarify, etc., depending on the requirements of the questions.

Reference Links of Dataset to download :

1. Youtube Video Dataset

[Youtube-Video-Views-Prediction/USvideos.csv at master · ZiHG/Youtube-Video-Views-Prediction · GitHub](#)

2. Salary Dataset

[Salary-Prediction-based-on-Years-of-Experience/Salary_Data.csv at master · sudarshankoirala/SalaryPrediction-based-on-Years-of-Experience · GitHub](#)

3. Heart Disease Dataset

<https://github.com/dileep-lingamallu/Heart-Disease-Prediction-Dataset/blob/main/heart.csv>

4. Online Gaming Dataset

[Online-Gaming/online_gaming_behavior_dataset.csv at main · PhuongFX/Online-Gaming · GitHub](#)

5. E-Commerce Customer Dataset

[customer-database/Ecommerce Customers.csv at master · araj2/customer-database · GitHub](#)

6. Spambase Dataset

https://github.com/ustunb/classificationpipeline/blob/master/Data/Raw%20Data%20Files/spam_base.csv

7. Sports Performance dataset

[Athlete_analysis/collegiate_athlete_injury_dataset.csv at main · cr7ritesh/Athlete_analysis · GitHub](#)

Duration: 3 Hours**Maximum Marks:35**

Q1. Create a dataframe containing information about different products such as Price, Rating, and Number of Sales. Generate a Dendrogram to group similar products based on their characteristics.

[10 Marks]

Q2. Create a suitable dataset named Loan_application.csv containing at least 10 records with the attributes Application_ID, Age, Income, Credit_Score, Loan_Amount, Employment_Years, Debt_Ratio, and Loan_Status, where Loan_Status represents whether the loan application is Approved or Rejected.

Write a Python program to perform the following data reduction techniques:

- a) Feature Selection: Identify the important features for predicting Loan_Status.
- b) Feature Extraction: Reduce the number of numerical attributes using a suitable technique such as Principal Component Analysis (PCA).

[20 Marks]

OR

Q2. Using the built-in Iris dataset, apply K-Means clustering with K = 3. Plot the datapoints, visualize the clusters with different colors, and display the cluster centroids.

[20 Marks]

Q3. Viva

[5 Mark]

Q1. Write a Python program to create an age_records DataFrame containing the attributes Person_ID and Age for suitable sample records. Perform Data Discretization on the Age attribute using Equal Width Binning. Display the original and discretized data.

[10 Marks]

Q2. Apply Linear Regression on a YouTube Video dataset to predict video views using relevant video attributes. Visualize the relationship between the predictor and target variable using a scatter plot and regression line.

[20 Marks]

OR

Q2. Create the following dataset in python. Apply the Apriori algorithm to generate frequent itemsets and association rules. Repeat the process with different minimum support values 0.2, and 0.3.

[20 Marks]

TID	Items
1	Video Lecture, Quiz
2	Video Lecture, Assignment
3	Quiz, Discussion Forum
4	Video Lecture, Quiz, Assignment
5	Assignment, Discussion Forum
6	Video Lecture, Certificate
7	Quiz, Certificate
8	Video Lecture, Quiz
9	Assignment, Certificate
10	Video Lecture, Quiz, Certificate

Q3. Viva

[5 Marks]

Q1. Create a dataframe containing information about different vehicles such as Fuel_Efficiency, Engine_Power, Vehicle_Weight, Engine_Size, and Acceleration. Add 5 records in dataframe, including outlier. Create Box Plots for Fuel Efficiency and Engine Power and identify outliers.

[10 Marks]

Q2. Consider the Salary Dataset. Write a Python program to read the dataset and display its information. Preprocess the data if required and split it into training and testing sets. Apply Simple Linear Regression to predict salary based on years of experience. Plot the regression line and evaluate the model using appropriate metrics such as Mean Squared Error.

[20 Marks]

OR

Q2. Apply the Random Forest algorithm on the Heart Disease dataset to predict the presence of heart disease. Preprocess the data, evaluate the model using accuracy and confusion matrix.

[20 Marks]

Q3. Viva

[5 Marks]

Q1. A survey was conducted to determine the preferred payment method among customers as shown below:

```
payment_methods = ["UPI", "Credit Card", "Debit Card", "Cash", "Net Banking"]
```

```
customers = [150, 80, 60, 40, 30]
```

Create a Pie Chart to represent the percentage of customers using each payment method. Display the percentage contribution of each category and add an appropriate title.

[10 Marks]

Q2. Apply Logistic Regression on an Online Gaming dataset to classify players as Casual or Professional gamers based on features such as daily play time, achievements unlocked, in-game purchases, and gaming level. Evaluate the model using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.

[20 Marks]

OR

Q2. Apply the K-Means Clustering algorithm on the given E-commerce Customer dataset. Read the dataset using Python, inspect and preprocess the data by handling missing values and selecting suitable numerical features. Apply K-Means clustering to segment the customers into different categories and interpret the resulting clusters.

[20 Marks]

Q3. Viva

[5 Marks]

Q1. Write a Python program to create a customer_DataFrame with suitable attributes such as Customer_ID, Gender, City, and Membership_Type. Perform label encoding technique on appropriate categorical attributes. Display the DataFrame before and after encoding.

[10 Marks]

Q2. Generate a DataFrame containing customer information for Netflix and Amazon Prime users. Create a Venn diagram and a Pie chart to show the customers using only Netflix, only Amazon Prime, and both services. Add appropriate labels, colors, and titles to both visualizations.

[20 Marks]

OR

Q2. Use the K-Nearest Neighbors (KNN) algorithm to classify emails as spam or non-spam using the Spambase dataset. Preprocess the data, train the KNN classifier, and evaluate its performance using accuracy, confusion matrix, precision, recall, and F1-score.

[20 Marks]

Q3. Viva

[5 Marks]

Q1. Create a DataFrame containing the car model, engine size (in litres), and fuel efficiency (in km/L) for the given cars. Create a scatter plot to visualize the relationship between engine size and fuel efficiency. Add an appropriate title, label both axes, and use different colors for the data points.

[10 Marks]

Q2. Use the Sports Performance dataset to develop a Simple Linear Regression model to predict Performance Score based on Training Hours Per Week. Evaluate the model using the R^2 score and plot a scatter graph with the regression line to visualize the relationship between training hours and performance score.

[20 Marks]

OR

Q2. Implement a KNN classifier on the given Customer dataset. Preprocess the data and classify customers into different segments. Evaluate the model using Accuracy and Confusion Matrix.

[20 Marks]

Q3. Viva

[5 Marks]

Q1. Consider the following dataset representing the number of students enrolled in different courses:

`courses = ["BCA", "BBA", "B.Com", "B.Sc", "BA"]`

`students = [60, 45, 75, 50, 35]`

Create a Bar Chart to visualize the number of students enrolled in each course. Use different bar colors, add an appropriate title, and label both axes.

[10 Marks]

Q2. Implement a KNN classifier on the given Student dataset. Preprocess the data and classify students into different performance categories. Evaluate the model using Accuracy and Confusion Matrix.

[20 Marks]

OR

Q2. Write a Python program to perform market basket analysis on the Groceries dataset using the Apriori algorithm. Read and explore the dataset, preprocess the data by handling missing values, convert categorical transaction data into a suitable numeric format, and generate frequent itemsets and association rules. Repeat the analysis using minimum support values 0.3 and 0.4.

[20 Marks]

S.No.	Items Purchased
1	Basmati Rice, Toor Dal, Turmeric Powder, Cooking Oil
2	Atta, Sugar, Tea, Milk
3	Rice, Moong Dal, Salt, Cooking Oil
4	Wheat Flour, Chickpeas, Turmeric Powder, Cumin Seeds
5	Poha, Peanuts, Onions, Cooking Oil
6	Basmati Rice, Paneer, Garam Masala, Tomatoes
7	Atta, Potatoes, Onions, Cooking Oil

Q3. Viva

[5 Marks]

Q1. Create a suitable dataset containing information about earthquakes using Latitude, Longitude, Depth, and Magnitude. Perform the following tasks:

- a. Create a Geospatial Visualization to display the locations of earthquakes on a map.
- b. Use color to represent the magnitude of each earthquake.

[10 Marks]

Q2. Create a suitable dataset named Loan_application.csv containing at least 10 records with the attributes Application_ID, Age, Income, Credit_Score, Loan_Amount, Employment_Years, Debt_Ratio, and Loan_Status, where Loan_Status represents whether the loan application is Approved or Rejected.

Write a Python program to perform the following data reduction techniques:

- a) Feature Selection: Identify the important features for predicting Loan_Status.
- b) Feature Extraction: Reduce the number of numerical attributes using a suitable technique such as Principal Component Analysis (PCA).

[20 Marks]

OR

Q2. Apply Simple Linear Regression on the Salary dataset to predict Salary using Years of Experience. Split the data into training and testing sets, plot the regression line, and evaluate the model using MSE.

[20 Marks]

Q3. Viva

[5 Marks]

Q1. Write a Python program to create a product DataFrame with the attributes Product_ID, Price, Quantity_Sold, Discount, and Revenue. Apply Min-Max Scaling data transformation techniques on the DataFrame and display the transformed data.

[10 Marks]

Q2. Apply Logistic Regression to classify smartphones into different price categories using specifications such as RAM, storage, battery capacity, and camera resolution. Assess the model using Accuracy, F1-Score, and Confusion Matrix.

[20 Marks]

OR

Q2. Create a dataset containing information about different books with the attributes Book Title, Author, Publication Year, Average Rating, and Number of Ratings.

Perform the following tasks on created dataset:

- a. Create a Venn Diagram comparing the authors of books published in two different time periods.
- b. Generate a Word Cloud using the book titles.

[20 Marks]

Q3. Viva

[5 Marks]

Q1. A company conducted a survey to determine the preferred mode of transportation among employees as shown below:

transport = ["Train", "Bicycle", "Bike", "Car", "Bus"],

employees = [220, 60, 70, 80, 40]

Create a Pie Chart to represent the percentage of employees using each mode of transportation.

Display the percentage contribution of each category. Give appropriate labels and colors.

[10 Marks]

Q2. Using the given housing dataset, develop a Random Forest Regression model to predict house prices based on the available property attributes. Perform the necessary data preprocessing, train and test the model, and evaluate its predictive performance using Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), and R^2 Score.

OR

[20 Marks]

Q2. Create the following dataset in python. Apply the Apriori algorithm to discover frequently used learning resources and generate association rules among them. Repeat the process with different minimum support values 0.3, and 0.4.

[20 Marks]

TID	Items
1	Video Lecture, Quiz
2	Video Lecture, Assignment
3	Quiz, Discussion Forum
4	Video Lecture, Quiz, Assignment
5	Assignment, Discussion Forum
6	Video Lecture, Certificate
7	Quiz, Certificate
8	Video Lecture, Quiz
9	Assignment, Certificate
10	Video Lecture, Quiz, Certificate

Q3. Viva

[5 Marks]

Q1. An organization has recorded the number of employees working in different departments such as HR, Finance, IT, Sales, and Marketing. Write a Python program to create a bar chart showing the distribution of employees among these departments. Display suitable labels and a title.

[10 Marks]

Q2. Apply Logistic Regression to classify smartphones into different price categories using the given smartphone dataset. Preprocess the data appropriately and assess the model using Accuracy, F1-Score, and Confusion Matrix.

[20 Marks]

OR

Q2. Apply the K-Means Clustering algorithm on the given E-commerce Customer dataset. Read the dataset using Python, inspect and preprocess the data by handling missing values and selecting suitable numerical features. Apply K-Means clustering to segment the customers into different categories and display the clusters.

[20 Marks]

Q3. Viva

[5 Marks]

Q1. Load the Titanic dataset (titanic.csv) Create a histogram showing the number of passengers in each passenger class. Create a Pie Chart showing the survival distribution.

[10 Marks]

Q2. Apply Polynomial Regression to predict song popularity using audio features such as danceability, energy, and tempo. Evaluate the model using MAE, RMSE, and R^2 Score, and visualize the fitted curve.

[20 Marks]

OR

Q2. Use the K-Nearest Neighbors (KNN) algorithm to classify emails as spam or non-spam using the Spambase dataset. Preprocess the data, train the KNN classifier, and evaluate its performance using accuracy, confusion matrix, precision, recall, and F1-score.

[20 Marks]

Q3. Viva

[5 Marks]

Q1. A retail store has recorded the daily sales for two months. Create a dataframe using sales.csv file and generate a Box Plot to identify any outliers. Customize the graph with title and labels.

[10 Marks]

Q2. Implement the K-Nearest Neighbors (KNN) algorithm to classify the Wine dataset. Perform the required data preprocessing, model training, and testing, and evaluate the model performance using Accuracy, Precision, Recall, and F1-Score.

[20 Marks]

OR

Q2. Apply LogisticRegression on an Online Gaming dataset to classify players as Casual or Professional gamers based on features such as daily play time, achievements unlocked, in-game purchases, and gaming level. Evaluate the model using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.

[20 Marks]

Q3. Viva

[5 Marks]

Q1. Create a dataframe containing information about different products such as Price, Rating, and Number of Sales. Generate a Dendrogram to group similar products based on their characteristics.

[10 Marks]

Q2. Using the Car Price Prediction dataset, apply Polynomial Regression to predict the Selling Price based on Vehicle Age, Present Price, and Kilometers Driven. Evaluate the model using the R^2 Score and visualize the actual data points along with the fitted polynomial curve.

[20 Marks]

OR

Q2. Create two DataFrames containing information about different mobile phones.

DataFrame 1: Brand, Model, Price, RAM

DataFrame 2: Model, Storage, Battery_Capacity, Operating_System

Perform the following data transformation techniques:

- a) Data Integration by combining the two DataFrames using Model
- b) Label Encoding on the Brand attribute of the merged DataFrame
- c) One-Hot Encoding on the Operating_System attribute of the merged DataFrame

Display the original DataFrames, merged DataFrame, and transformed DataFrame.

[20 Marks]

Q3. Viva

[5 Marks]

Q1. Write a Python program to create an age_records DataFrame containing the attributes Person_ID and Age for suitable sample records. Perform Data Discretization on the Age attribute using Equal Frequency Binning. Display the original and discretized data.

[10 Marks]

Q2. Create a DataFrame containing information about different cities with the attributes City, Area (sq. km), Average Temperature (°C), Rainfall (mm), and Population Density. Generate a Treemap to visualize the population density of each city and create a 3D Scatter Plot using Area, Average Temperature, and Rainfall. Add suitable labels, colors, and titles to both visualizations.

[20 Marks]

OR

Q2. Write a Python program to perform market basket analysis on the Groceries dataset using the Apriori algorithm. Read and explore the dataset, preprocess the data by handling missing values, convert categorical transaction data into a suitable numeric format, and generate frequent itemsets and association rules. Repeat the analysis using minimum support values 0.3 and 0.4.

[20 Marks]

S.No.	Items Purchased
1	Basmati Rice, Toor Dal, Turmeric Powder, Cooking Oil
2	Atta, Sugar, Tea, Milk
3	Rice, Moong Dal, Salt, Cooking Oil
4	Wheat Flour, Chickpeas, Turmeric Powder, Cumin Seeds
5	Poha, Peanuts, Onions, Cooking Oil
6	Basmati Rice, Paneer, Garam Masala, Tomatoes
7	Atta, Potatoes, Onions, Cooking Oil

Q3. Viva

[5 Marks]

Q1. Create a DataFrame containing student performance data with attributes such as Study Hours, Attendance, Assignment Score, and Exam Score. Calculate the correlation matrix for these attributes and generate a Heat Map to visualize the correlation between them. Display the correlation values, suitable labels, and an appropriate title.

[10 Marks]

Q2. Write a Python program to create a DataFrame with the attributes Temperature, Humidity, Wind-Speed, Pressure, and Rainfall. Apply the following data transformation techniques and display the transformed data:

- a)Min-MaxScaling
- b)Standardization
- c) Normalization

[20 Marks]

OR

Q2. Implement the Apriori algorithm on the given student activity dataset to identify frequent itemsets and generate association rules among the activities. Repeat the analysis using minimum support values 0.2 and 0.3.

[20 Marks]

TID	Items
1.	Library, Seminar
2.	Workshop, Project
3.	Library, Assignment
4.	Seminar, Workshop
5.	Assignment, Seminar
6.	Workshop, Assignment
7.	Project, Library
8.	Library, Seminar, Project
9.	Library,Seminar,Workshop

Q3. Viva

[5 Marks]

Q1. Write a Python program to create two DataFrames, student_details and academic_records, containing suitable student-related attributes. Perform data integration by combining the two DataFrames using a common attribute such as Student_ID. Display the individual DataFrames and the integrated DataFrame.

[10 Marks]

Q2. Create a dataset containing information about different cities with the attributes: City, Population, GDP, and Literacy Rate. Perform the following visualization tasks:

- a. Generate a Treemap showing city-wise population distribution.
- b. Create a 3D Scatter Plot using Population, GDP, and Literacy Rate.

[20 Marks]

OR

Q2. Develop a Random Forest Classifier using the given Student Grading dataset to classify students into different performance categories. Perform the necessary data preprocessing, model training, and testing, and evaluate the classifier using appropriate classification metrics such as Confusion matrix, Accuracy, Precision, Recall, and F1-Score.

[20 Marks]

Q3. Viva

[5 Marks]

Duration: 3 Hours

Maximum Marks:35

Q1. Write a Python program to create an employee_data DataFrame with the attributes: Emp_ID, Age, Dept, Years_Experience, and Performance_Rating. Perform data cleaning by handling missing values, remove duplicate records and detecting outliers using the IQR method.

[10 Marks]

Q2. Consider the World Cities dataset containing attributes - city, country, latitude, longitude, and population information. Perform the following tasks:

- a. Create a Geospatial Visualization showing the locations of cities on a world map, using marker size to represent population.
- b. Create a Treemap to visualize the population distribution of cities.

[20 Marks]

OR

Q2. Use the Crop Yield dataset to apply Simple Linear Regression for predicting crop yield from rainfall. Handle missing values, split the data into training and testing sets, and evaluate the model using MAE and R^2 Score. Create a Scatter Plot with a Regression Line to visualize the relationship.

[20 Marks]

Q3. Viva

[5 Marks]

Duration: 3 Hours**Maximum Marks:35**

Q1. Create a dataframe containing information about different vehicles such as Fuel_Efficiency, Engine_Power, Vehicle_Weight, Engine_Size, and Acceleration. Add 5 records in dataframe, including outlier. Create Box Plots for Fuel Efficiency and Engine Power and identify outliers.

[10 Marks]

Q2. Write a Python program to create a DataFrame with the attributes Temperature, Humidity, Wind Speed, Pressure, and Rainfall. Apply the following data transformation techniques and display the transformed data:

a)Min-MaxScaling

b)Standardization

c) Normalization

[20 Marks]

OR

Q2. Implement the Random Forest Classification algorithm on the given Heart Disease dataset to predict the presence or absence of heart disease. Perform the necessary data preprocessing, model training, and testing, and evaluate the model using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.

[20 Marks]

Q3. Viva

[5 Marks]

Q1. Write a Python program to create a DataFrame containing the attributes Age, Salary, Experience, Working_Hours, and Performance_Score. Apply the Standardization data transformation technique and display the transformed data.

[10 Marks]

Q2. Create a dataset containing information about different books with the attributes Book Title, Author, Publication Year, Average Rating, and Number of Ratings.

Perform the following tasks on created dataset:

- Create a Venn Diagram comparing the authors of books published in two different time periods.
- Generate a Word Cloud using the book titles.

[20 Marks]

OR

Q2. Apply the Apriori algorithm to the following student activity dataset to identify frequent itemsets and generate association rules among the activities. Repeat the analysis using minimum support values 0.2 and 0.3.

[20 Marks]

TID	Items
1.	Library , Seminar
2.	Workshop, Project
3.	Library, Assignment
4.	Seminar, Workshop
5.	Assignment, Seminar
6.	Workshop, Assignment
7.	Project, Library
8.	Library, Seminar, Project
9.	Library,Seminar,Workshop

Q3. Viva

[5 Marks]

Q1. Create a dataframe using weather.csv containing values of Temperature, Humidity, Rainfall, and Wind Speed. Generate a Heat Map to visualize the correlation between these weather parameters.

[10 Marks]

Q2. Apply the K-Means Clustering algorithm to the given student dataset using Attendance and Marks as the input attributes. Perform the required data preprocessing, set $K = 3$, assign the students to the respective clusters, and visualize the clusters and their centroids using a suitable scatter plot.

[20 Marks]

OR

Q2. Create a suitable dataset named Loan_application.csv containing at least 10 records with the attributes Application_ID, Age, Income, Credit_Score, Loan_Amount, Employment_Years, Debt_Ratio, and Loan_Status, where Loan_Status represents whether the loan application is Approved or Rejected.

Write a Python program to perform the following data reduction techniques:

- a) Feature Selection: Identify the important features for predicting Loan_Status.
- b) Feature Extraction: Reduce the number of numerical attributes using a suitable technique such as Principal Component Analysis (PCA).

[20 Marks]

Q3. Viva

[5 Marks]

Q1. Write a Python program to create a DataFrame containing the attributes City, House_Type, House_Area, Number_of_Rooms, and Monthly_Rent. Apply the Normalization technique to the numerical attributes and display the normalized data.

[10 Marks]

Q2. Generate a DataFrame containing customer information for Netflix and Amazon Prime users. Create a Venn diagram and a Pie chart to show the customers using only Netflix, only Amazon Prime, and both services. Add appropriate labels, colors, and titles to both visualizations.

[20 Marks]

OR

Q2. Apply the K-Means Clustering algorithm to a student dataset containing Attendance and Marks. Perform the necessary data preprocessing, use $K = 3$ to group the students into three clusters, and visualize the resulting clusters along with their cluster centroids using a scatter plot.

[20 Marks]

Q3. Viva

[5 Marks]

Duration: 3 Hours**Maximum Marks:35**

Q1 Create a DataFrame containing information about different products with attributes such as Product_name, Price, Rating, and Number_of_Sales. Generate a Dendrogram to identify groups of similar products. Display a suitable title and labels for the Dendrogram.

[10 Marks]

Q2. Write a Python program to create a DataFrame with the attributes Temperature, Humidity, Wind Speed, Pressure, and Rainfall. Apply the following data transformation techniques and display the transformed data:

- a)Min-MaxScaling
- b)Standardization
- c) Normalization

[20 Marks]

OR

Q2. Implement a K-Nearest Neighbors (KNN) classifier on the given Student Grading dataset to classify students into different performance categories. Perform the necessary data preprocessing, train and test the model, and evaluate its performance using Accuracy, Precision, Recall, F1-Score, and Confusion Matrix.

[20 Marks]

Q3. Viva

[5 Marks]

Q1. Write a Python program to create a book_records DataFrame containing the attributes Book_ID and Book_Price for suitable sample records. Perform Data Discretization on the Book_Price attribute using Equal Width Binning. Display the original and discretized data.

[10 Marks]

Q2. Consider the World Cities dataset containing attributes city, country, latitude, longitude, and population information. Perform the following visualization tasks :

- a. Create a Geospatial Visualization showing the locations of cities on a world map, using marker size to represent population.
- b. Create a Treemap to visualize the population distribution of cities.

[20 Marks]

OR

Q2. Consider the Car Price Prediction dataset and apply Polynomial Regression to predict the Selling Price based on Vehicle Age, Present Price, and Kilometers Driven. Perform the necessary data preprocessing, train and test the model, evaluate its performance using the R^2 Score, and visualize the actual data points along with the fitted polynomial regression curve.

[20 Marks]

Q3. Viva

[5 Marks]

Savitribai Phule Pune University
T.Y.B.Sc.(C.S.) NEP – 2020 Practical Examination
Lab Course CS-308-MJ-P
Data Science and Analytics

Slip No: 25

Duration: 3 Hours

Maximum Marks:35

Q1. Consider the following dataset representing the number of students enrolled in different courses:
courses = ["BCA", "BBA", "B.Com", "B.Sc", "BA"]
students = [60, 45, 75, 50, 35]
Create a Bar Chart to visualize the number of students enrolled in each course. Use different bar colors, add an appropriate title, and label both axes.

[10 Marks]

Q2. Create two DataFrames containing information about different mobile phones.
DataFrame 1: Brand, Model, Price, RAM
DataFrame 2: Model, Storage, Battery_Capacity, Operating_System

Perform the following data transformation techniques:

- a) Data Integration by combining the two DataFrames using Model
 - b) Label Encoding on the Brand attribute of the merged DataFrame
 - c) One-Hot Encoding on the Operating_System attribute of the merged DataFrame
- Display the original DataFrames, merged DataFrame, and transformed DataFrame.

[20 Marks]

OR

Q2. Write a Python program to perform market basket analysis on the Groceries dataset using the Apriori algorithm. Read and explore the dataset, preprocess the data by handling missing values, convert categorical transaction data into a suitable numeric format, and generate frequent itemsets and association rules. Repeat the analysis using minimum support values 0.3 and 0.4.

[20 Marks]

S.No.	Items Purchased
1	Basmati Rice, Toor Dal, Turmeric Powder, Cooking Oil
2	Atta, Sugar, Tea, Milk
3	Rice, Moong Dal, Salt, Cooking Oil
4	Wheat Flour, Chickpeas, Turmeric Powder, Cumin Seeds
5	Poha, Peanuts, Onions, Cooking Oil
6	Basmati Rice, Paneer, Garam Masala, Tomatoes
7	Atta, Potatoes, Onions, Cooking Oil

Q3. Viva

[5 Marks]